Project Executable Files

Date	04 July 2026
Team ID	[Enter Team ID]
Project Name	HDI Predictor
Maximum Marks	3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide	Yes
3	requirements.txt	Yes
4	Database / raw dataset (data/hdi_dataset.csv)	Yes
5	Saved Model files (model/scaler.pkl, model/best_model.pkl)	Yes
6	Dockerfile	Yes
7	Notebook plots (EDA images)	Yes

Step 2: File / Folder Structure

```
HDI_Predictor/
■■■ data/
■ ■■■ hdi_dataset.csv # UNDP historical dataset
■■■ model/
■ ■■■ best_model.pkl # Serialized best ML model (joblib)
■ ■■■ scaler.pkl # Serialized fitted StandardScaler (joblib)
■■■ notebooks/
■ ■■■ plots/ # Generated EDA scatter/heatmap/trend plots
■■■ static/
■ ■■■ style.css # Glassmorphism application stylesheet
■■■ templates/
■ ■■■ index.html # Inputs form view
■ ■■■ result.html # Prediction outputs view
■■■ train_model.py # ML wrangling, training & comparison pipeline
■■■ app.py # Flask web application router
■■■ Dockerfile # Deployment instructions container
■■■ requirements.txt # Python packaging dependencies
■■■ README.md # Project main user instructions
```

Step 3: Deployment / Access Details

Hosting Provider: Render / Hugging Face Spaces (using Docker)
Local Run Link: http://127.0.0.1:5000

Step 4: Local Run Instructions

1. Put the raw **hdi_dataset.csv** inside the **data/** directory.
2. Execute **python train_model.py** to auto-compare, select, and save the best model and scaler.
3. Start the Flask application by running **python app.py**.
4. Open the browser and visit **http://127.0.0.1:5000** to test predictions.